Lab Exercise 17- Creating Secrets in Kubernetes

Creating secrets in Kubernetes is an essential practice for managing sensitive information such as passwords, OAuth tokens, and SSH keys. Let's go through a lab exercise that demonstrates how to create different types of secrets in Kubernetes.

Step 1: Create a Generic Secret

Create a secret from literal values:

```
kubectl create secret generic my-secret --from-literal=username=admin --from-literal=password='P@ssword'
```

```
C:\Users\shwet>kubectl create secret generic my-secret --from-literal=username=admin --from-literal=password=P@ssw0rd
secret/my-secret created
```

Verify the secret:

```
kubectl get secrets my-secret -o yaml
```

```
C:\Users\shwet>kubectl get secrets my-secret -o yaml
apiVersion: v1
data:
  password: UEBzc3cwcmQ=
  username: YWRtaW4=
kind: Secret
metadata:
  creationTimestamp: "2026-04-07T18:22:39Z"
  name: my-secret
  namespace: default
  resourceVersion: "26889"
  uid: 12536462-9e58-47ef-b4aa-8f083003c855
type: Opaque
```

The output will be base64 encoded. To decode, you can use:

```
echo -n “P@ssword” | base64
```

Step 2: Create a Secret from a File

Create a file with your sensitive data:

```
echo -n 'my-username' > ./username.txt



echo -n 'my-password' > ./password.txt
```

Create a secret from the file:

```
kubectl create secret generic my-file-secret --from-file=username=./username.txt --from-file=password=./password.txt
```

```
C:\Users\shwet>kubectl create secret generic my-file-secret --from-file=username=username.txt --from-file=password=passw
ord.txt
secret/my-file-secret created
```

Verify the secret:

```
kubectl get secrets my-file-secret -o yaml
```

```
C:\Users\shwet>kubectl get secrets my-file-secret -o yaml
apiVersion: v1
data:
  password: bXktcGFzc3dvcmQ=
  username: bXktdXNlcm5hbWU=
kind: Secret
metadata:
  creationTimestamp: "2026-04-07T18:26:06Z"
  name: my-file-secret
  namespace: default
  resourceVersion: "27165"
  uid: b5b57bfd-c566-4f9d-8a13-c55b1d003f20
type: Opaque
```

Step 3: Create a Docker Registry Secret

Create a Docker registry secret:

kubectl create secret docker-registry my-registry-secret --docker-username=<your-username> --docker-password=<your-password> --docker-email=<your-email> --docker-server=<your-docker-server>

```
C:\Users\shwet>kubectl create secret docker-registry my-registry-secret --docker-username=user --docker-password=pass --
docker-email=email@gmail.com --docker-server=https://index.docker.io/v1/
secret/my-registry-secret created
```

Verify the secret:

kubectl get secrets my-registry-secret -o yaml

```
C:\Users\shwet>kubectl get secrets my-registry-secret -o yaml
apiVersion: v1
data:
  .dockerconfigjson: eyJhdXRocyI6eyJodHRwczovL2luZGV4LmRvY2tlci5pby92MS8iOnsidXNlcm5hbWUiOiJ1c2VyIiwicGFzc3dvcmQiOiJwYXN
zIiwiZW1haWwiOiJlbWFpbEBnbWFpbC5jb20iLCJhdXRoIjoiZFhObGNqcHdZWE56In19fQ==
kind: Secret
metadata:
  creationTimestamp: "2026-04-07T18:28:17Z"
  name: my-registry-secret
  namespace: default
  resourceVersion: "27335"
  uid: 42d60b1f-2b00-452c-81e9-6778a047c6a9
type: kubernetes.io/dockerconfigjson
```

Step 4: Create a Secret Using a YAML Manifest

Create a YAML file for the secret:

Create a file named **my-secret.yaml** with the following content:

```
apiVersion: v1
kind: Secret
metadata:
  name: my-manifest-secret
type: Opaque
data:
  username: YWRtaW4=    # base64 encoded value of 'admin'
  password: UEBzc3cwcmQ=  # base64 encoded value of 'P@ssw0rd'
```

Apply the YAML file:

```
kubectl apply -f my-manifest-secret.yaml
```

```
C:\Users\shwet>kubectl apply -f pod-with-secret.yaml
secret/my-manifest-secret created
```

Verify the secret:

```
kubectl get secrets my-manifest-secret -o yaml
```

```
C:\Users\shwet>kubectl get secrets my-manifest-secret -o yaml
apiVersion: v1
data:
  password: UEBzc3cwcmQ=
  username: YWRtaW4=
kind: Secret
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"v1","data":{"password":"UEBzc3cwcmQ=","username":"YWRtaW4="},"kind":"Secret","metadata":{"annotatio
ns":{},"name":"my-manifest-secret","namespace":"default"},"type":"Opaque"}
  creationTimestamp: "2026-04-07T18:30:24Z"
  name: my-manifest-secret
  namespace: default
  resourceVersion: "27504"
  uid: 06952d80-e870-41f7-9bdd-2bd473333e3d
type: Opaque
```

Step 5: Use Secrets in Pods

Create a pod that uses the secret:

Create a file named **pod-with-secret.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: pod-with-secret
spec:
  containers:
  - name: mycontainer
    image: nginx
    env:
    - name: USERNAME
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: username
    - name: PASSWORD
      valueFrom:
        secretKeyRef:
          name: my-secret
```

```
      key: password
```

Apply the YAML file:

```
kubectl apply -f pod-with-secret.yaml
```

```
C:\Users\shwet>notepad pod-with-secret.yaml

C:\Users\shwet>kubectl apply -f pod-with-secret.yaml
pod/pod-with-secret created
```

Verify the pod is running and the secret is being used:

```
kubectl exec pod-with-secret -- printenv | grep -E 'USERNAME|PASSWORD'
```

```
C:\Users\shwet>kubectl exec pod-with-secret -- printenv
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=pod-with-secret
USERNAME=admin
PASSWORD=P@ssw0rd
NODEPORT_SERVICE_PORT=tcp://10.98.155.180:80
NODEPORT_SERVICE_PORT_80_TCP_PORT=80
NODEPORT_SERVICE_PORT_80_TCP_ADDR=10.98.155.180
KUBERNETES_SERVICE_HOST=10.96.0.1
KUBERNETES_PORT_443_TCP_ADDR=10.96.0.1
NODEPORT_SERVICE_SERVICE_HOST=10.98.155.180
NODEPORT_SERVICE_SERVICE_PORT=80
NODEPORT_SERVICE_PORT_80_TCP_PROTO=tcp
KUBERNETES_SERVICE_PORT=443
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT_443_TCP=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP_PORT=443
NODEPORT_SERVICE_PORT_80_TCP=tcp://10.98.155.180:80
KUBERNETES_PORT=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP_PROTO=tcp
NGINX_VERSION=1.29.7
```

Step 7: Clean Up

Delete the secrets:

kubectl delete secret my-secret

kubectl delete secret my-file-secret

kubectl delete secret my-registry-secret

kubectl delete secret my-manifest-secret

```
C:\Users\shwet>kubectl delete secret my-secret
secret "my-secret" deleted

C:\Users\shwet>kubectl delete secret my-file-secret
secret "my-file-secret" deleted

C:\Users\shwet>kubectl delete secret my-manifest-secret
secret "my-manifest-secret" deleted

C:\Users\shwet>kubectl delete pod pod-with-secret
pod "pod-with-secret" deleted
```